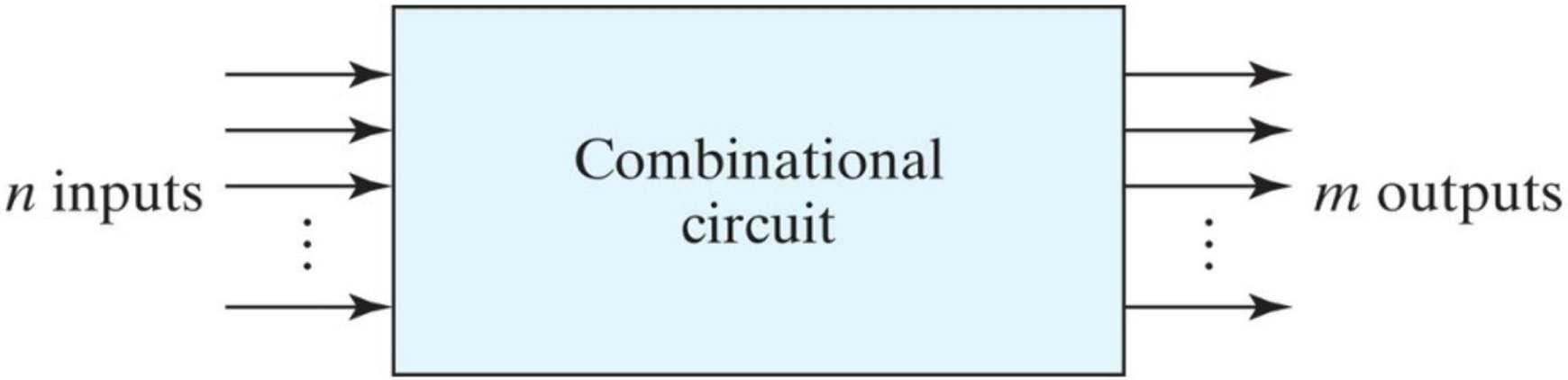
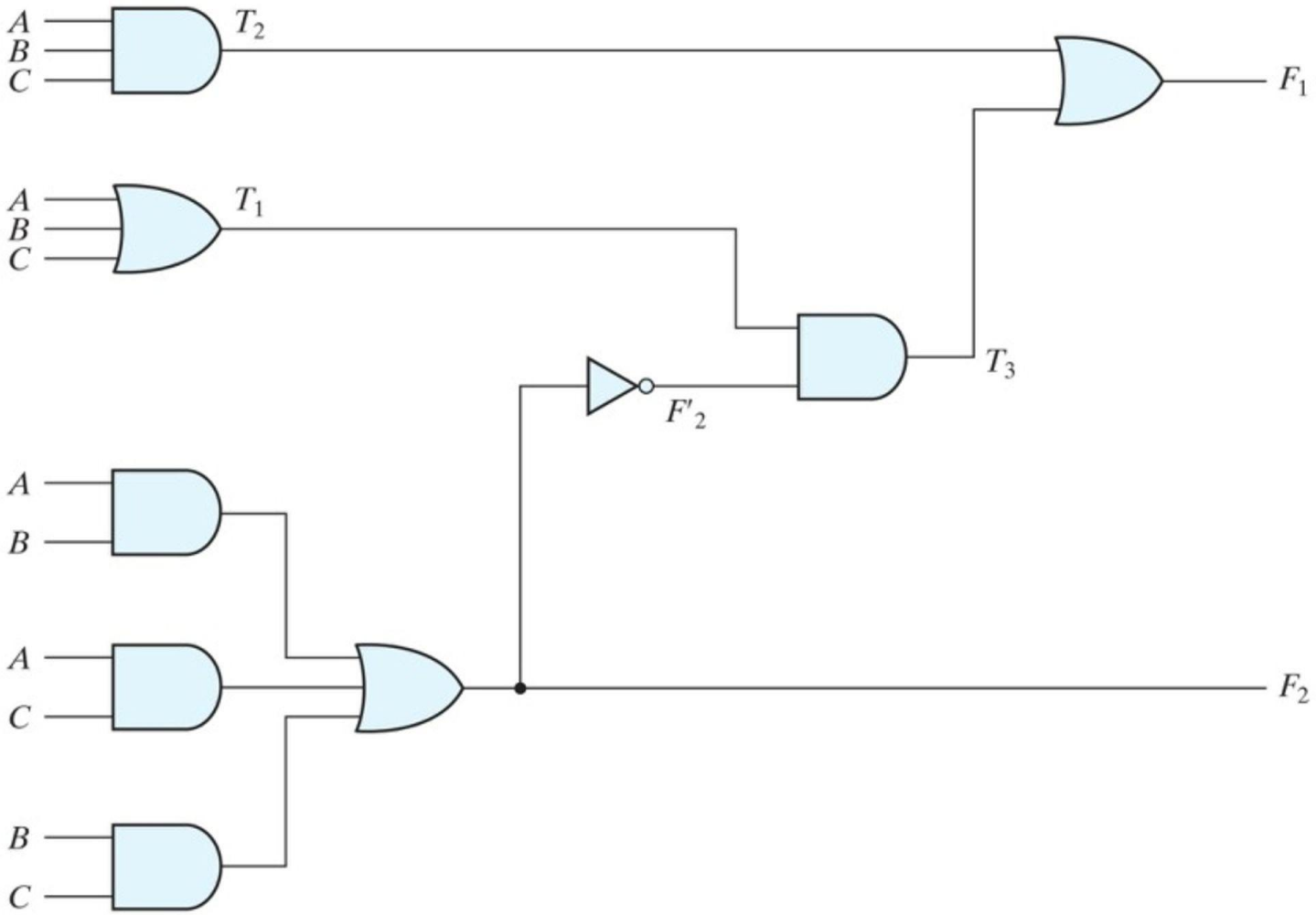
FIGURE 4.1 Block diagram of combinational circuit



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FIGURE 4.2 Logic diagram for analysis example



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Table 4.1 Truth Table for the Logic Diagram of Fig. 4.2

Table 4.1
Truth Table for the Logic Diagram of Fig. 4.2

A	B	C	F_2	F'_2	T_1	T_2	T_3	F_1
0	0	0	0	1	0	0	0	0
0	0	1	0	1	1	0	1	1
0	1	0	0	1	1	0	1	1
0	1	1	1	0	1	0	0	0
1	0	0	0	1	1	0	1	1
1	0	1	1	0	1	0	0	0
1	1	0	1	0	1	0	0	0
1	1	1	1	0	1	1	0	1

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Table 4.2 Truth Table for Code Conversion Example

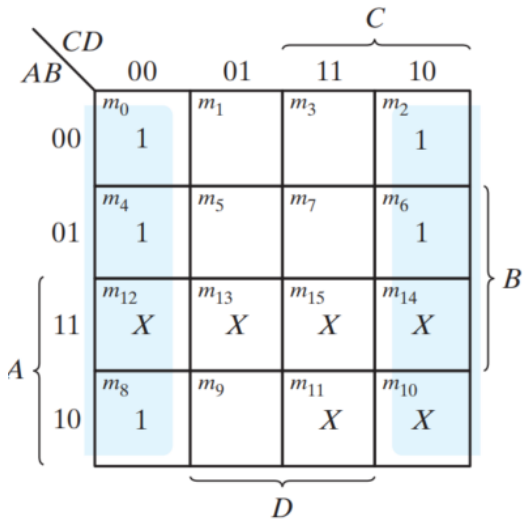
Table 4.2
Truth Table for Code Conversion Example

Input BCD				Output Excess-3 Code			
A	*B*	*C*	*D*	*w*	*x*	*y*	*z*
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0

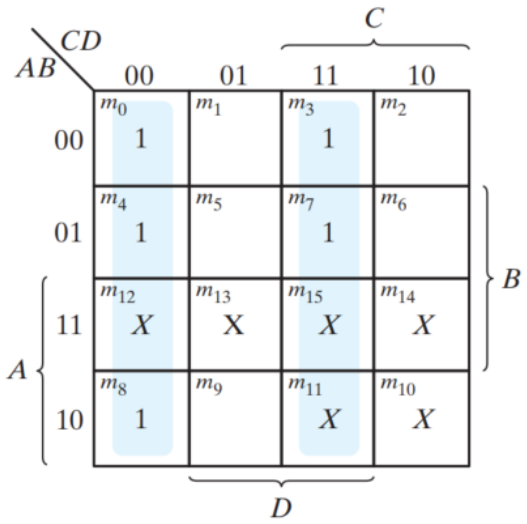
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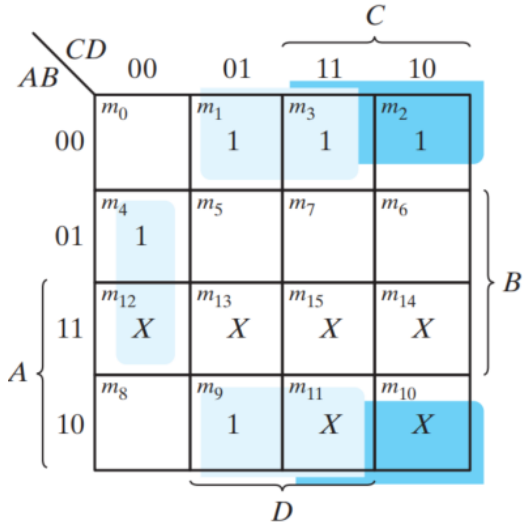
FIGURE 4.3 Maps for BCD-to-excess-3 code converter



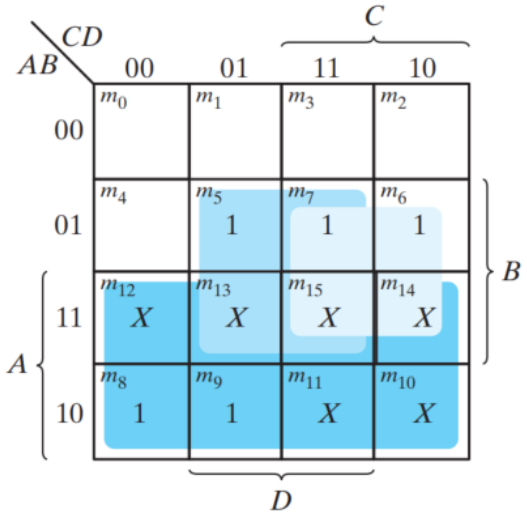
$z = D'$



$y = CD + C'D'$



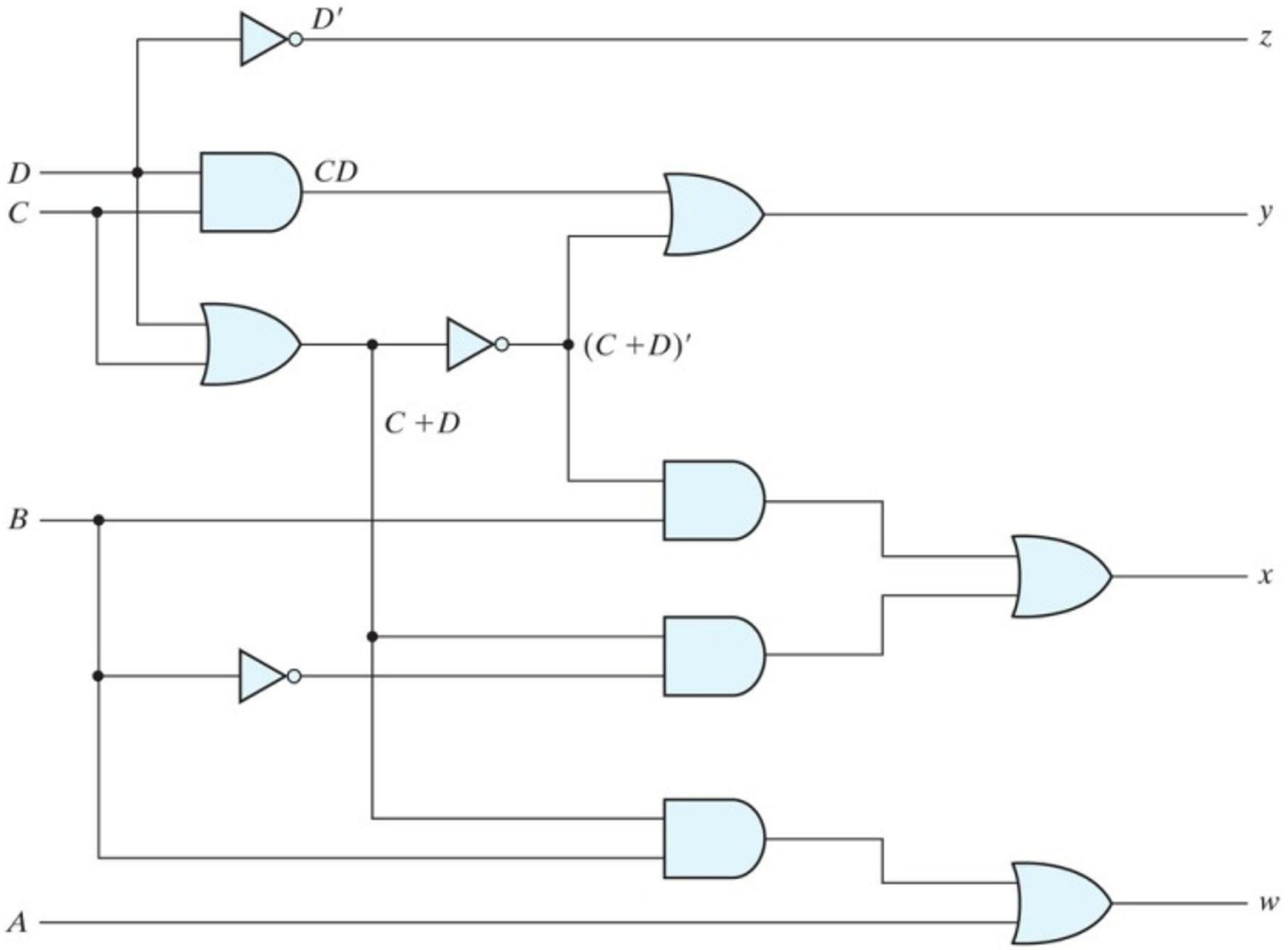
$x = B'C + B'D + BC'D'$



$w = A + BC + BD$

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FIGURE 4.4 Logic diagram for BCD-to-excess-3 code converter



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Table 4.3 Half Adder

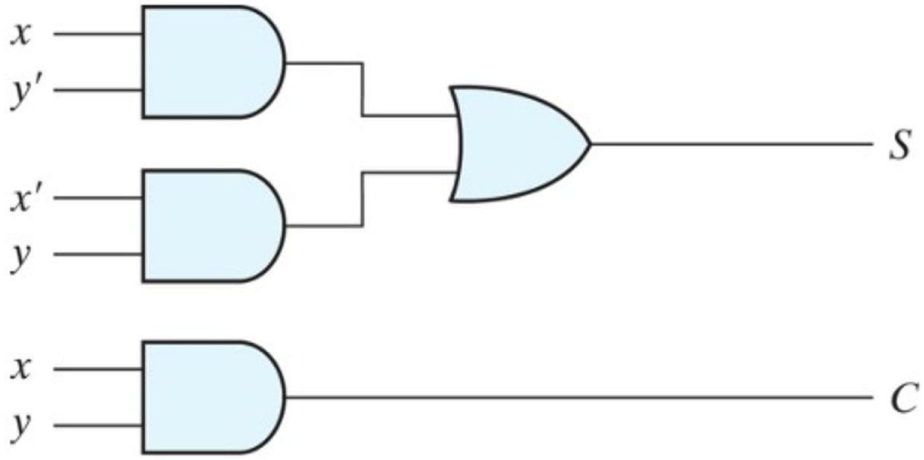
Table 4.3
Half Adder

x	*y*	*C*	*S*
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

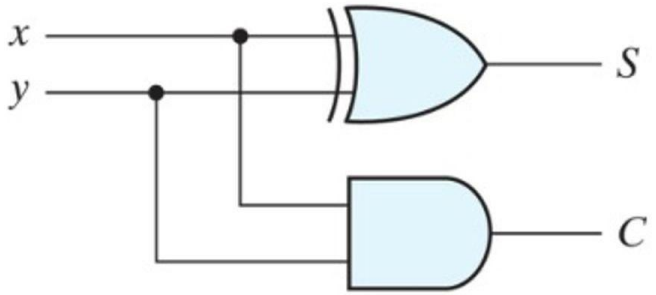
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FIGURE 4.5 Implementation of half adder



(a) $S = xy' + x'y$
$C = xy$



(b) $S = x \oplus y$
$C = xy$

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Table 4.4 Full Adder

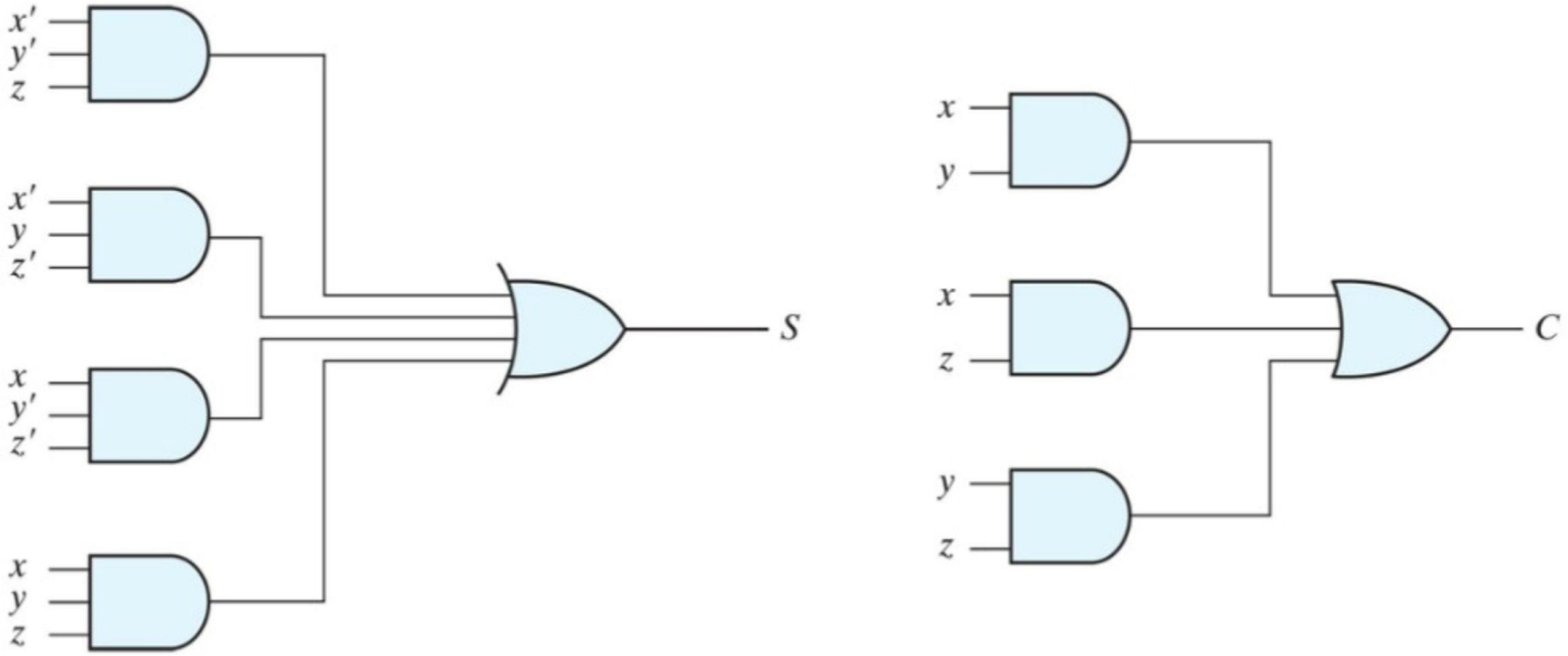
Table 4.4
Full Adder

x	*y*	*z*	*C*	*S*
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

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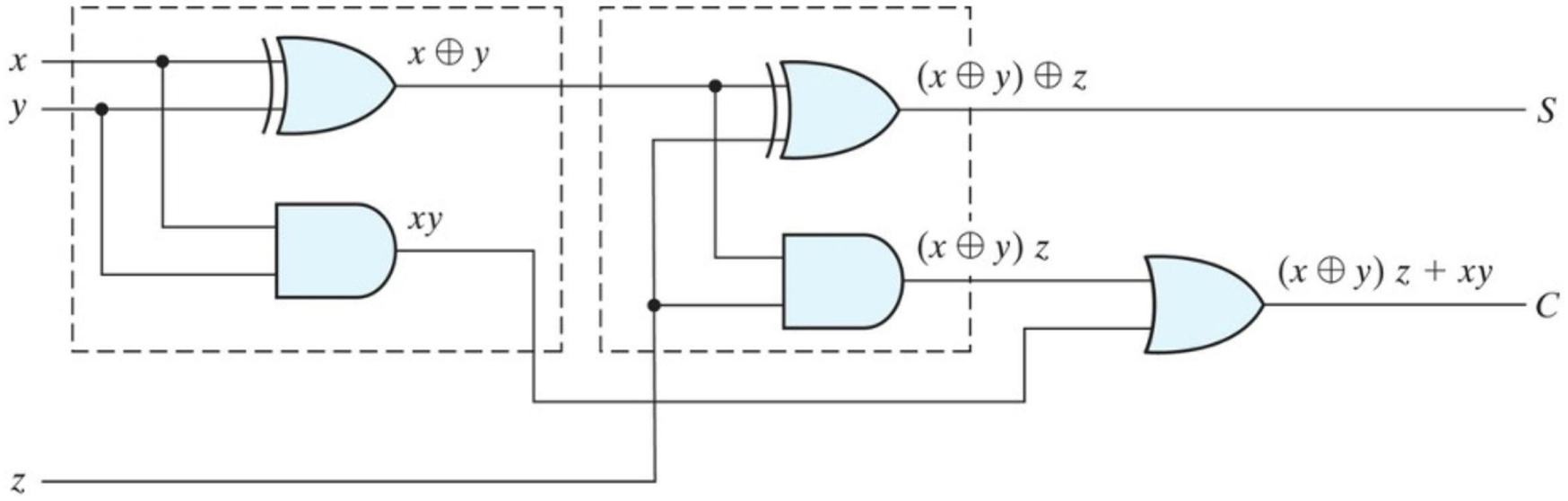
FIGURE 4.7 Implementation of full adder in sum-of-products form



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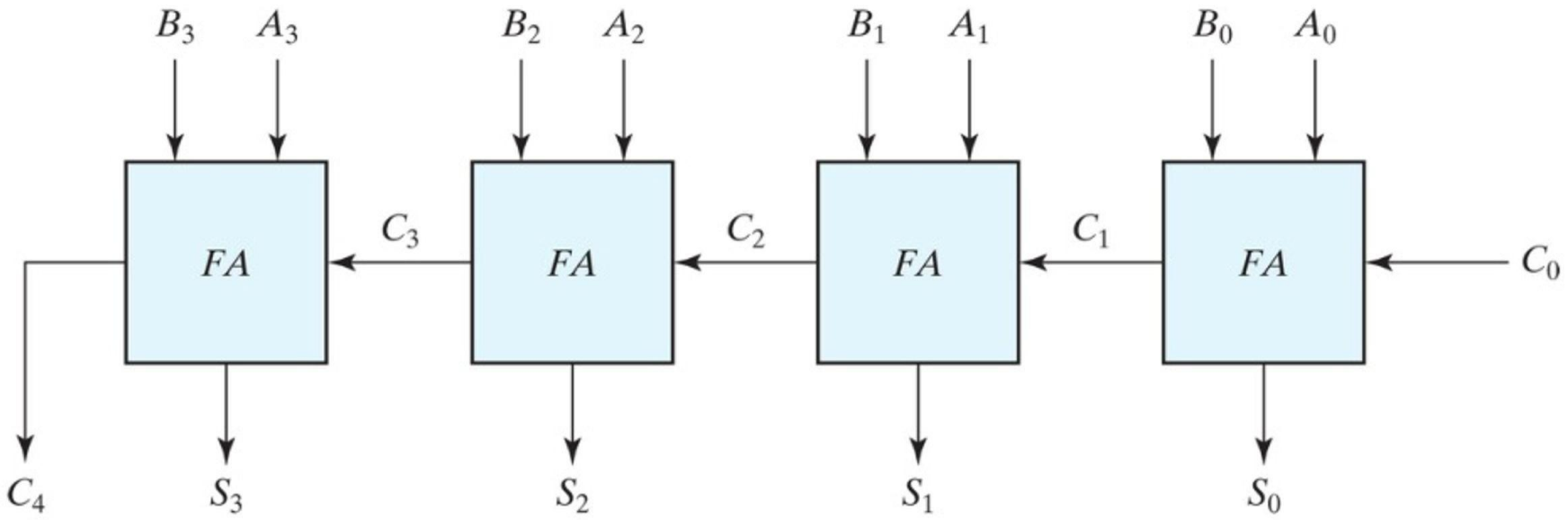
FIGURE 4.8 Implementation of full adder with two half adders and an OR gate



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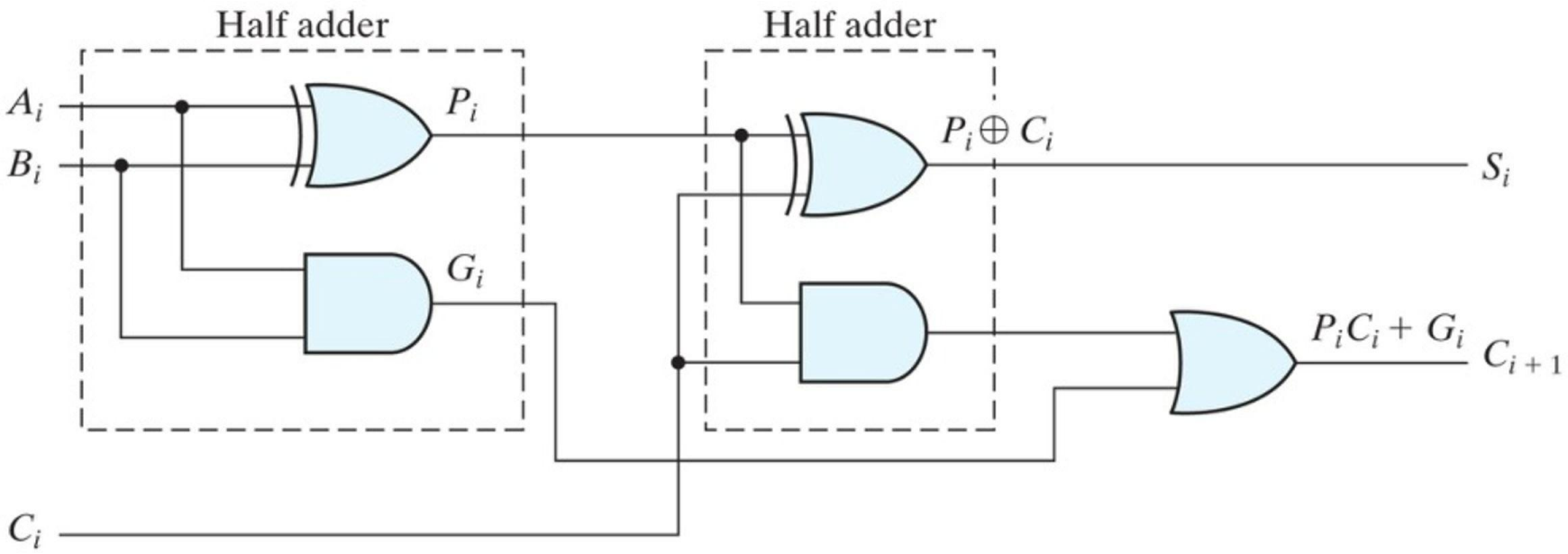
FIGURE 4.9 Four-bit adder



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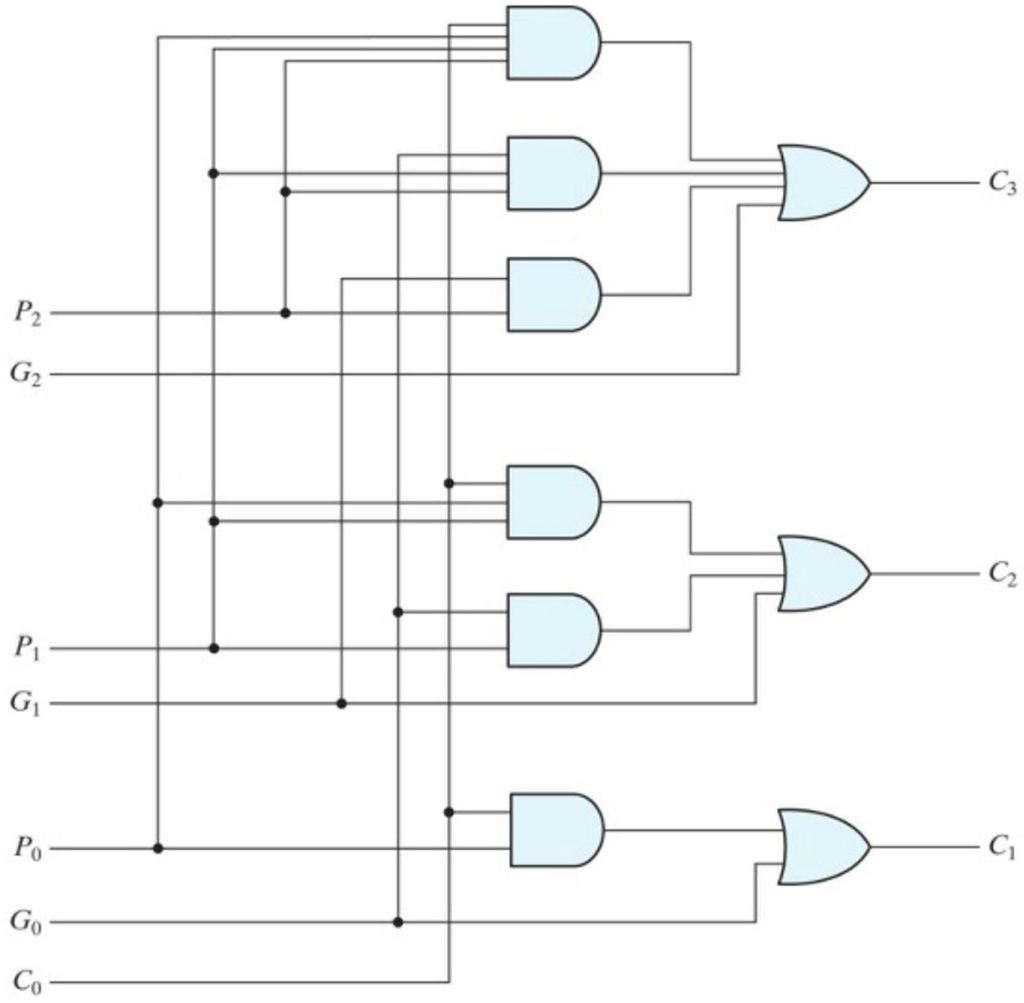
FIGURE 4.10 Full adder with *P* and *G* shown



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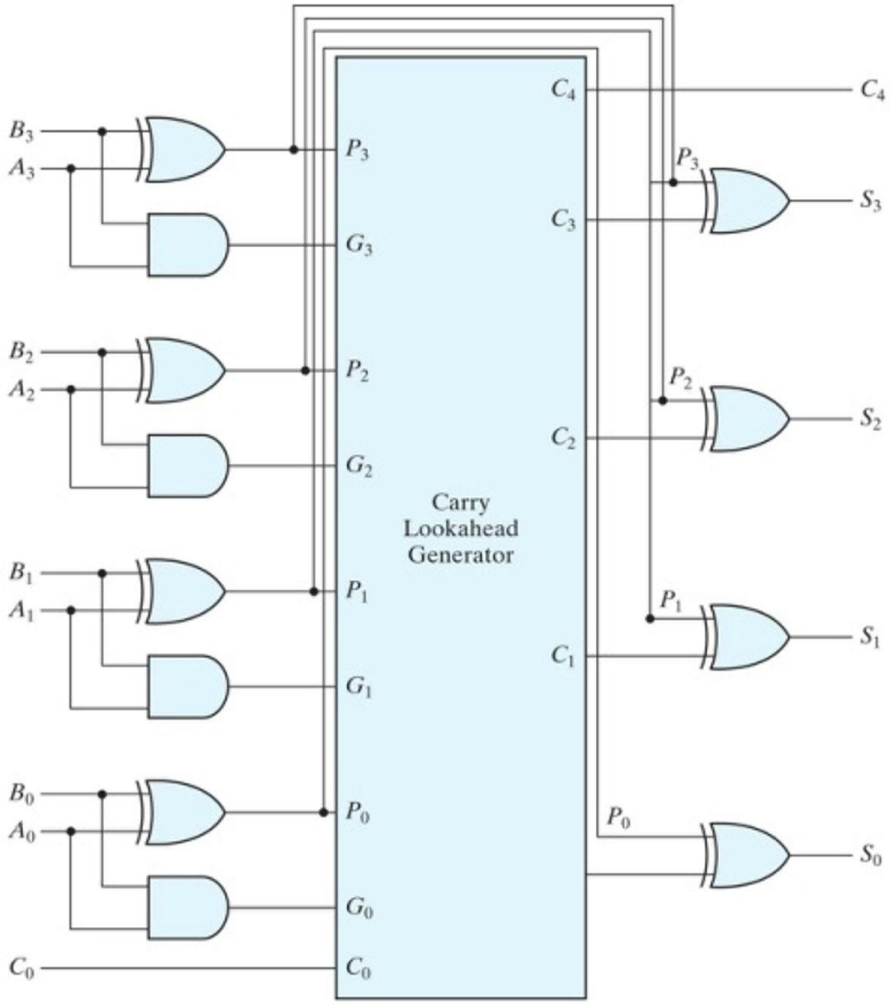
FIGURE 4.11 Logic diagram of carry lookahead generator



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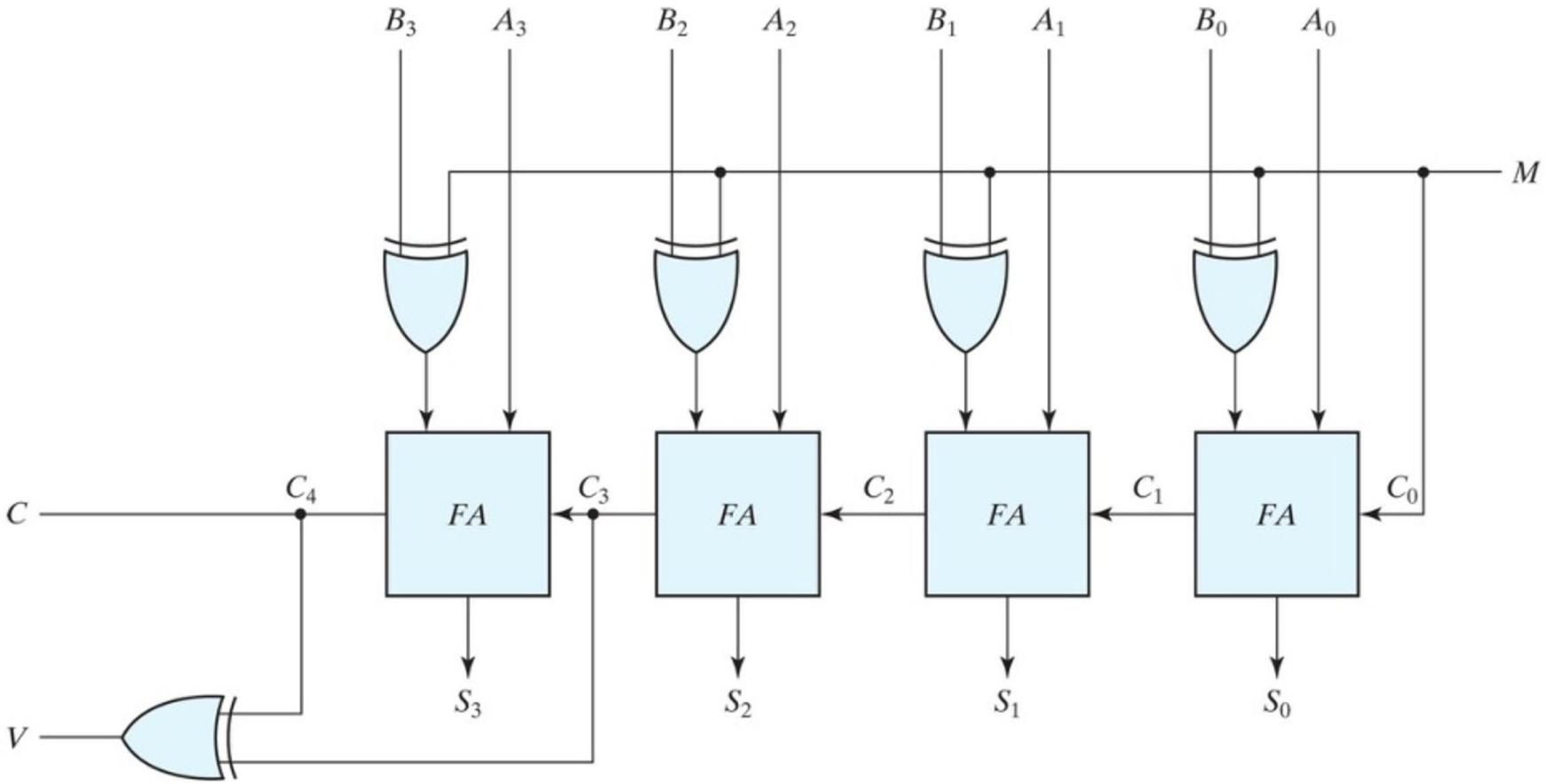
FIGURE 4.12 Four-bit adder with carry lookahead



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FIGURE 4.13 Four-bit adder–subtractor (with overflow detection)



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Table 4.5 Derivation of BCD Adder

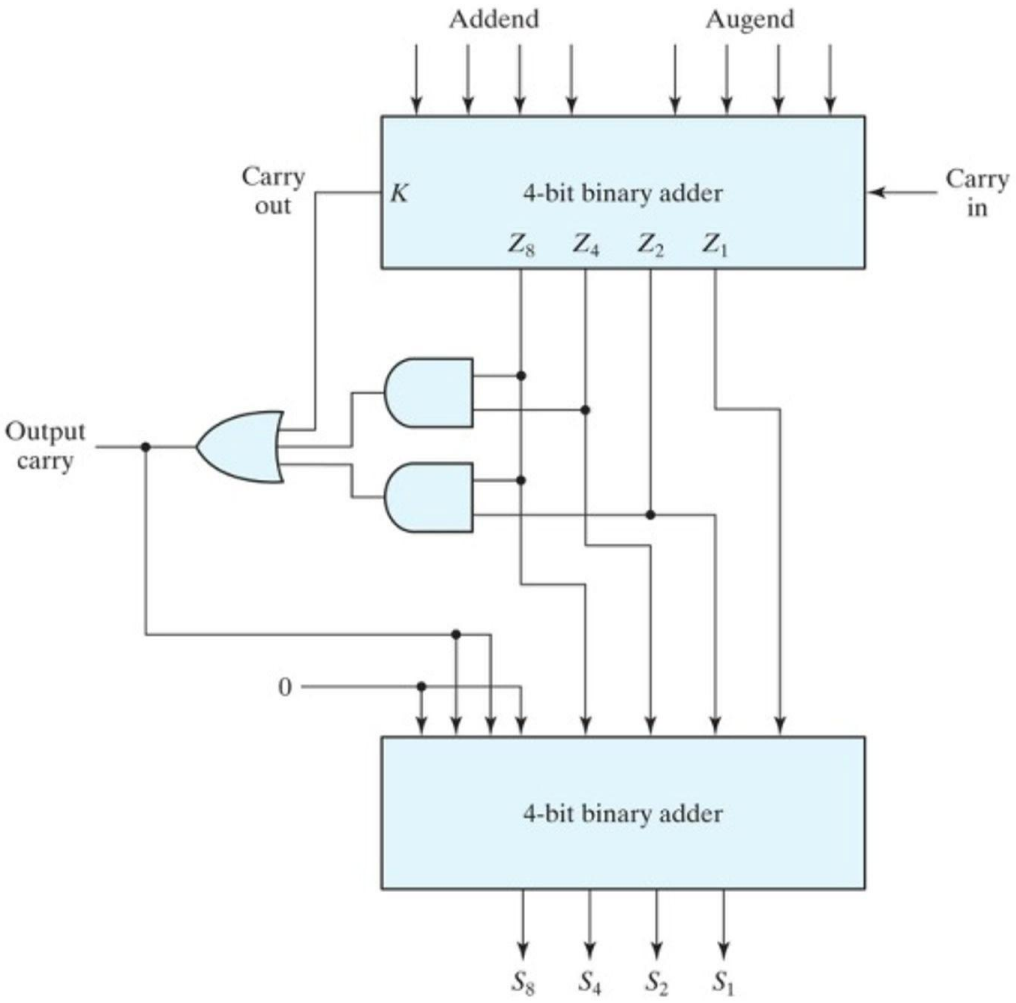
Table 4.5
Derivation of BCD Adder

Binary Sum					BCD Sum					Decimal
K	Z_8	Z_4	Z_2	Z_1	C	S_8	S_4	S_2	S_1	
0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	2
0	0	0	1	1	0	0	0	1	1	3
0	0	1	0	0	0	0	1	0	0	4
0	0	1	0	1	0	0	1	0	1	5
0	0	1	1	0	0	0	1	1	0	6
0	0	1	1	1	0	0	1	1	1	7
0	1	0	0	0	0	1	0	0	0	8
0	1	0	0	1	0	1	0	0	1	9
0	1	0	1	0	1	0	0	0	0	10
0	1	0	1	1	1	0	0	0	1	11
0	1	1	0	0	1	0	0	1	0	12
0	1	1	0	1	1	0	0	1	1	13
0	1	1	1	0	1	0	1	0	0	14
0	1	1	1	1	1	0	1	0	1	15
1	0	0	0	0	1	0	1	1	0	16
1	0	0	0	1	1	0	1	1	1	17
1	0	0	1	0	1	1	0	0	0	18
1	0	0	1	1	1	1	0	0	1	19

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FIGURE 4.14 Block diagram of a BCD adder

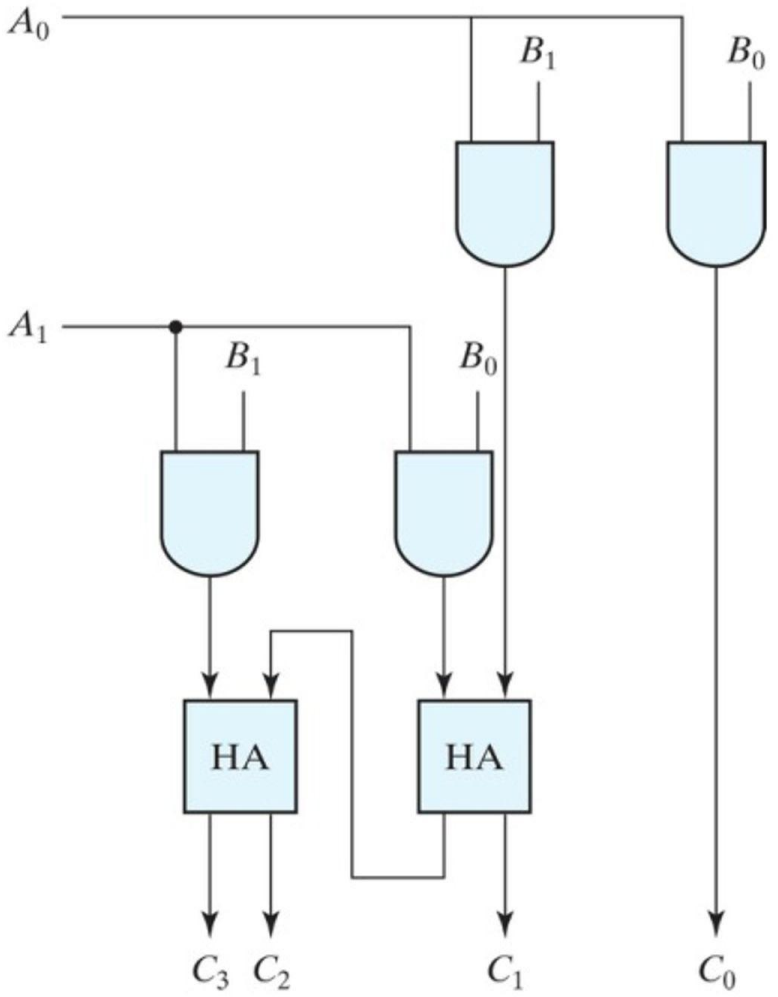


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FIGURE 4.15 Two-bit by two-bit binary multiplier

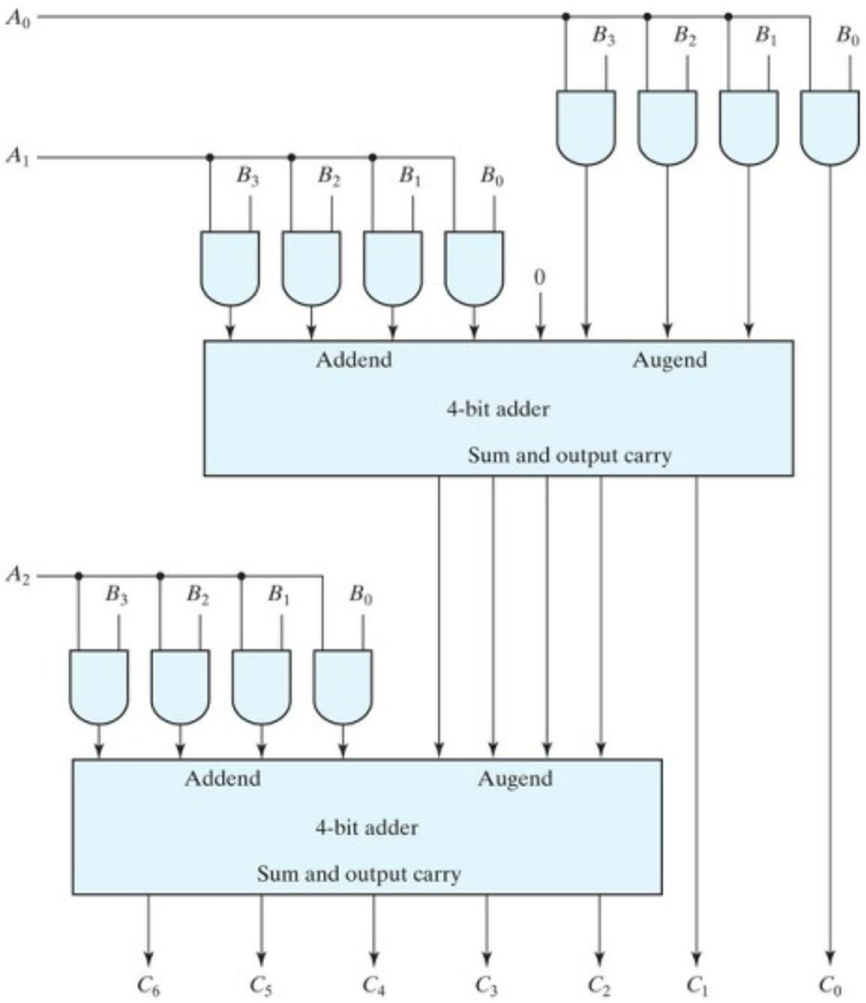
$$
\begin{array}{cccc}
 & & B_1 & B_0 \\
 & & A_1 & A_0 \\
\hline
 & & A_0B_1 & A_0B_0 \\
 & A_1B_1 & A_1B_0 & \\
\hline
C_3 & C_2 & C_1 & C_0
\end{array}
$$



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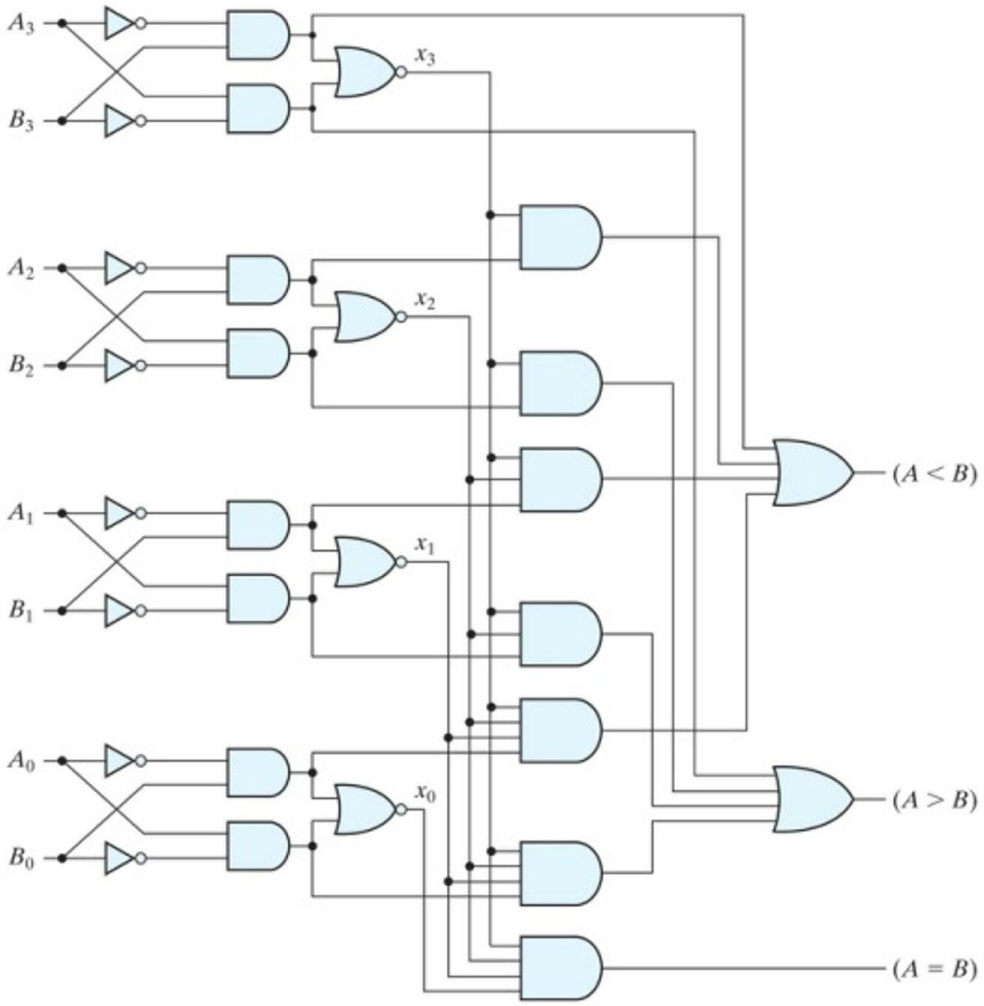
FIGURE 4.16 Four-bit by three-bit binary multiplier



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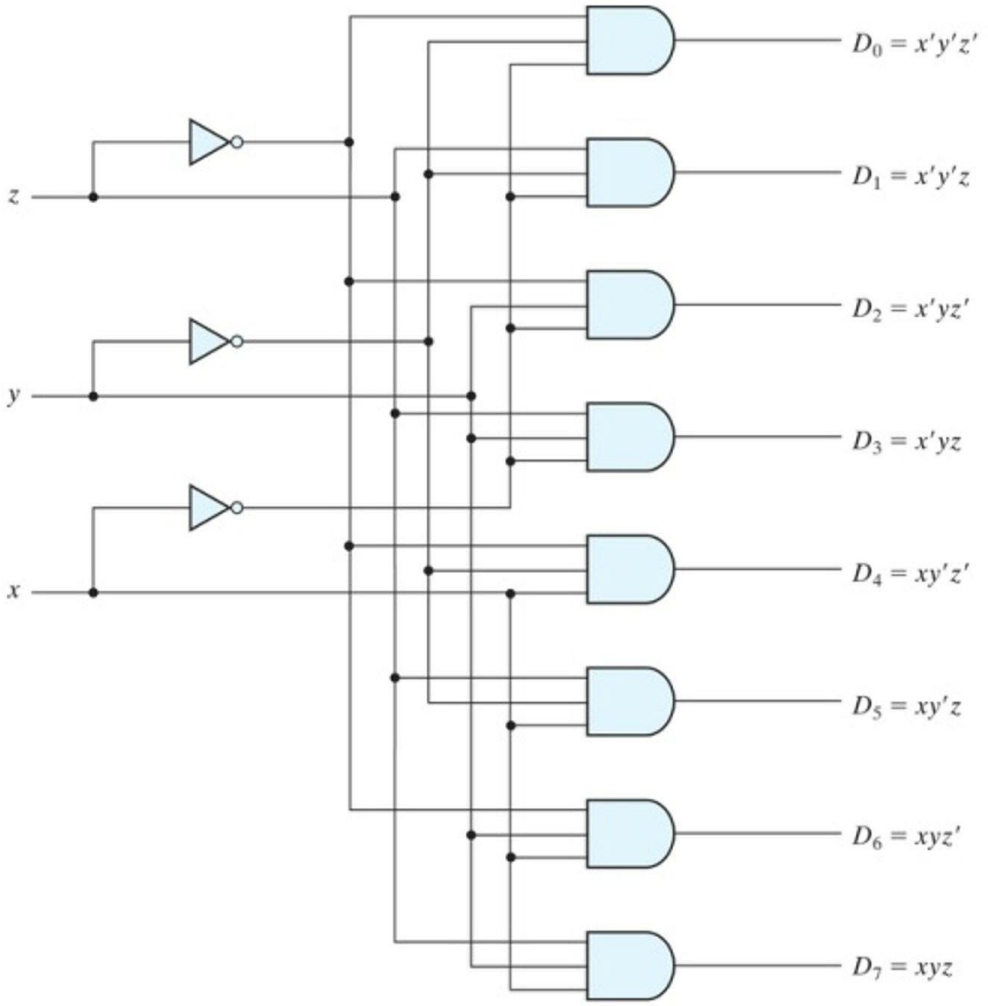
FIGURE 4.17 Four-bit magnitude comparator



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FIGURE 4.18 Three-to-eight-line decoder



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Table 4.6 Truth Table of a Three-to-Eight-Line Decoder

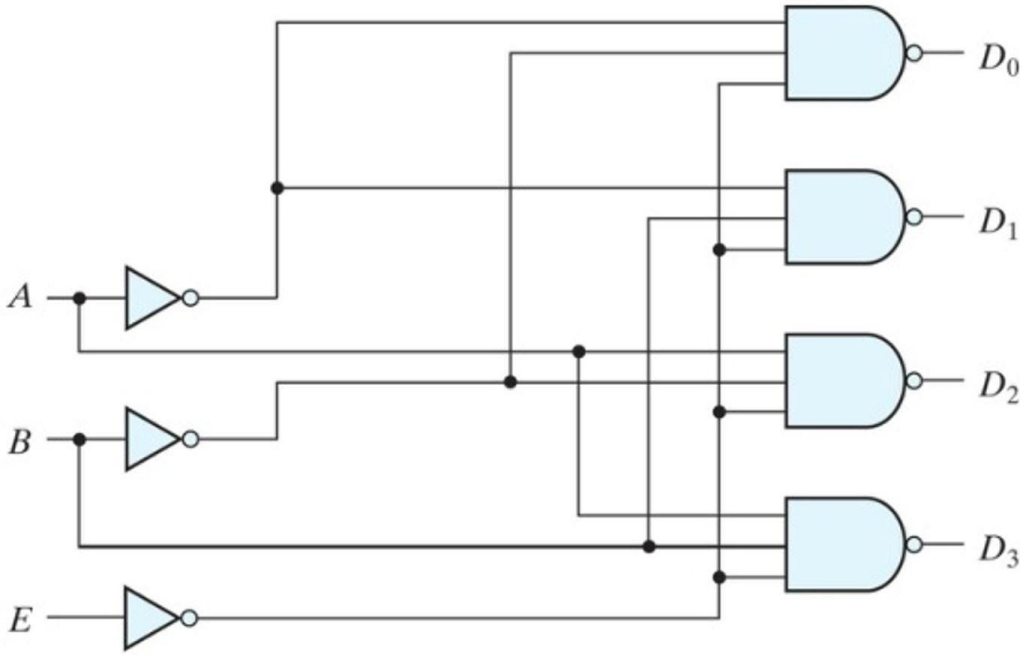
Table 4.6
Truth Table of a Three-to-Eight-Line Decoder

Inputs			Outputs							
x	y	z	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

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FIGURE 4.19 Two-to-four-line decoder with enable input



(a) Logic diagram

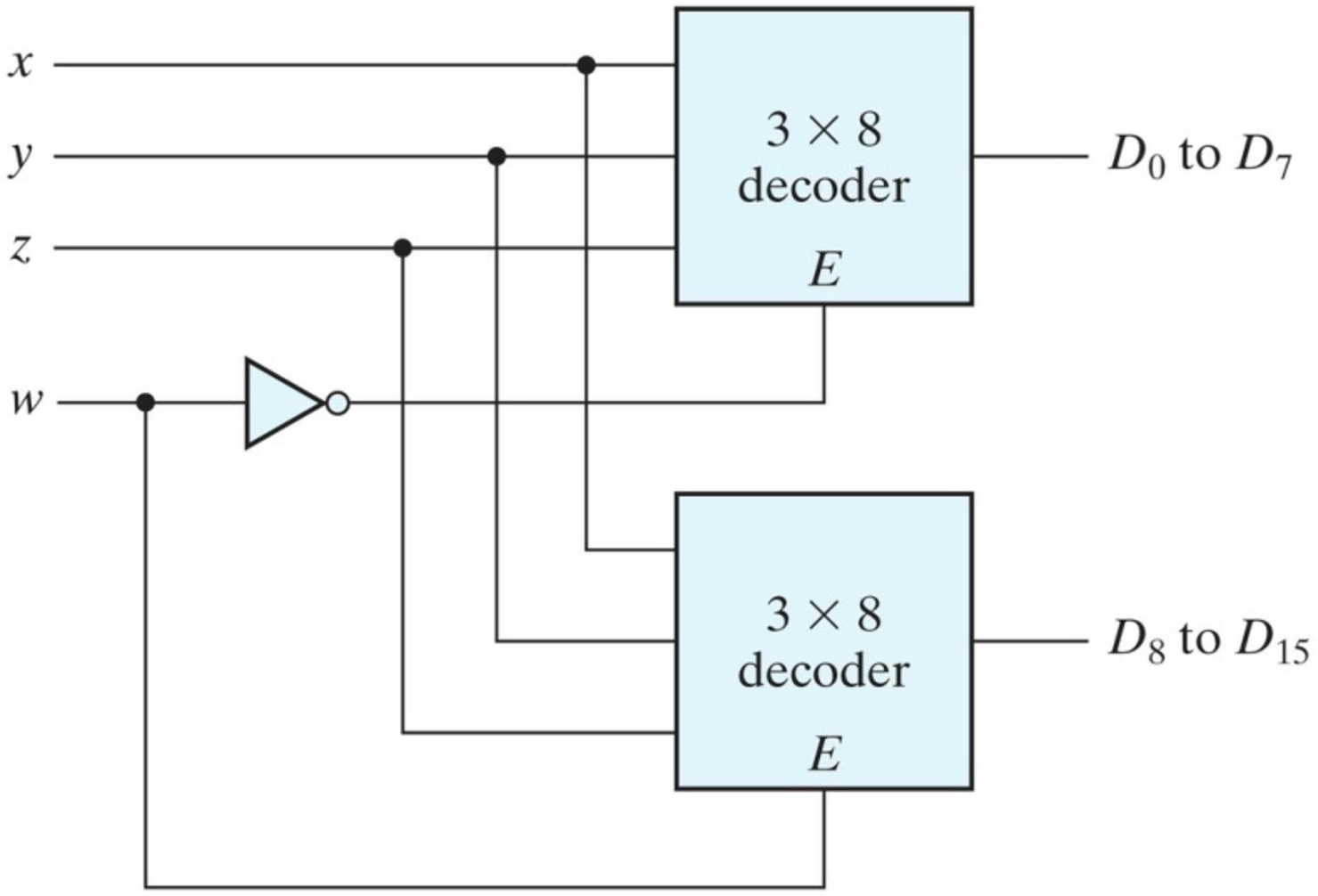
E	A	B	D_0	D_1	D_2	D_3
1	X	X	1	1	1	1
0	0	0	0	1	1	1
0	0	1	1	0	1	1
0	1	0	1	1	0	1
0	1	1	1	1	1	1

(b) Truth table

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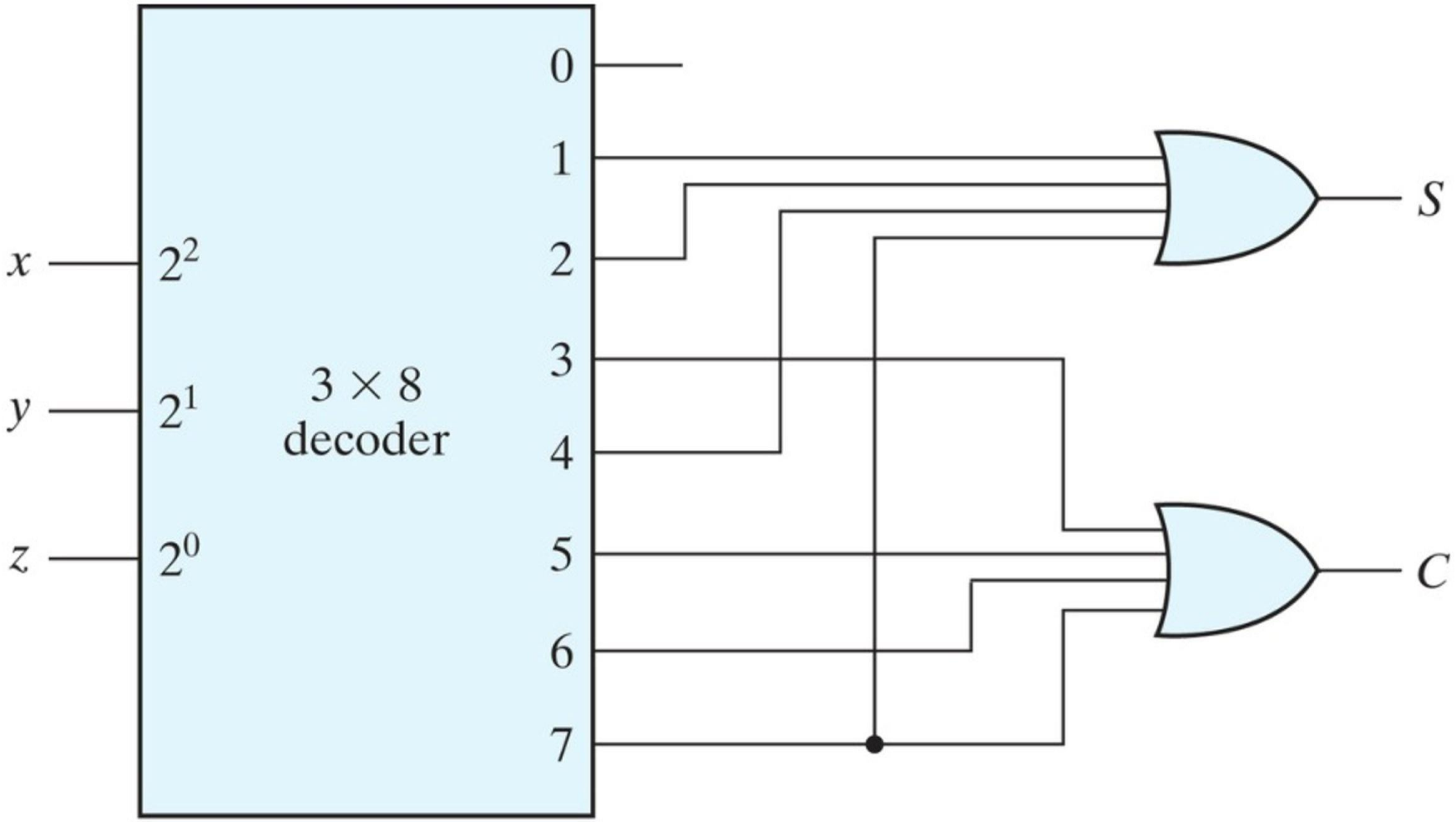
FIGURE 4.20 4 × 16 decoder constructed with two 3 × 8 decoders



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FIGURE 4.21 Implementation of a full adder with a decoder



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Table 4.7 Truth Table of an Octal-to-Binary Encoder

Table 4.7
Truth Table of an Octal-to-Binary Encoder

Inputs								Outputs		
D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	x	y	z
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

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Table 4.8 Truth Table of a Priority Encoder

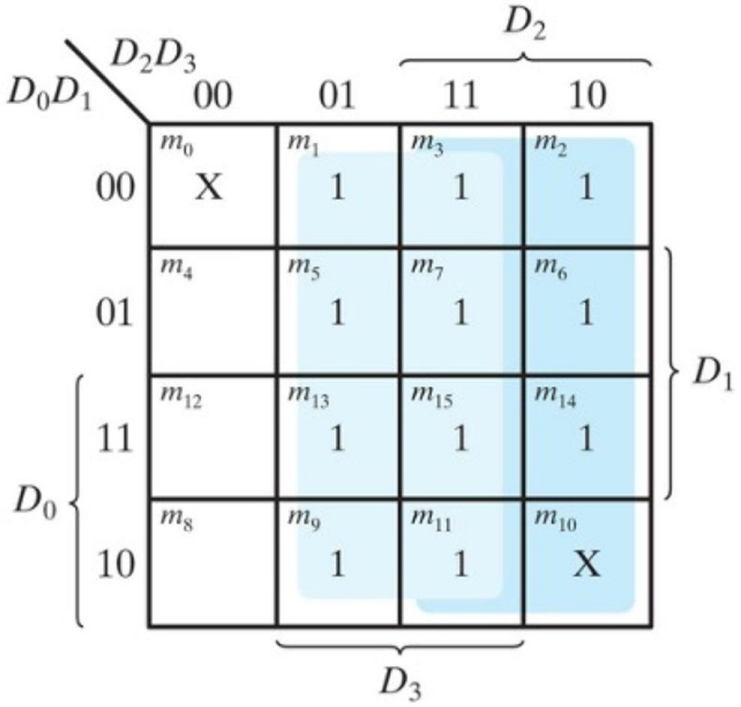
Table 4.8
Truth Table of a Priority Encoder

Inputs				Outputs		
D_0	D_1	D_2	D_3	x	y	V
0	0	0	0	X	X	0
1	0	0	0	0	0	1
X	1	0	0	0	1	1
X	X	1	0	1	0	1
X	X	X	1	1	1	1

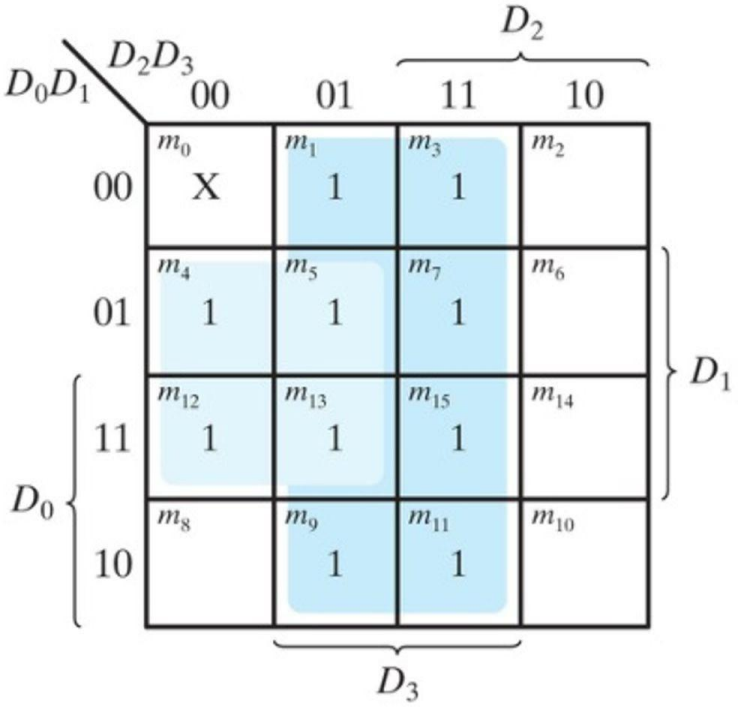
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FIGURE 4.22 Maps for a priority encoder



$x = D_2 + D_3$

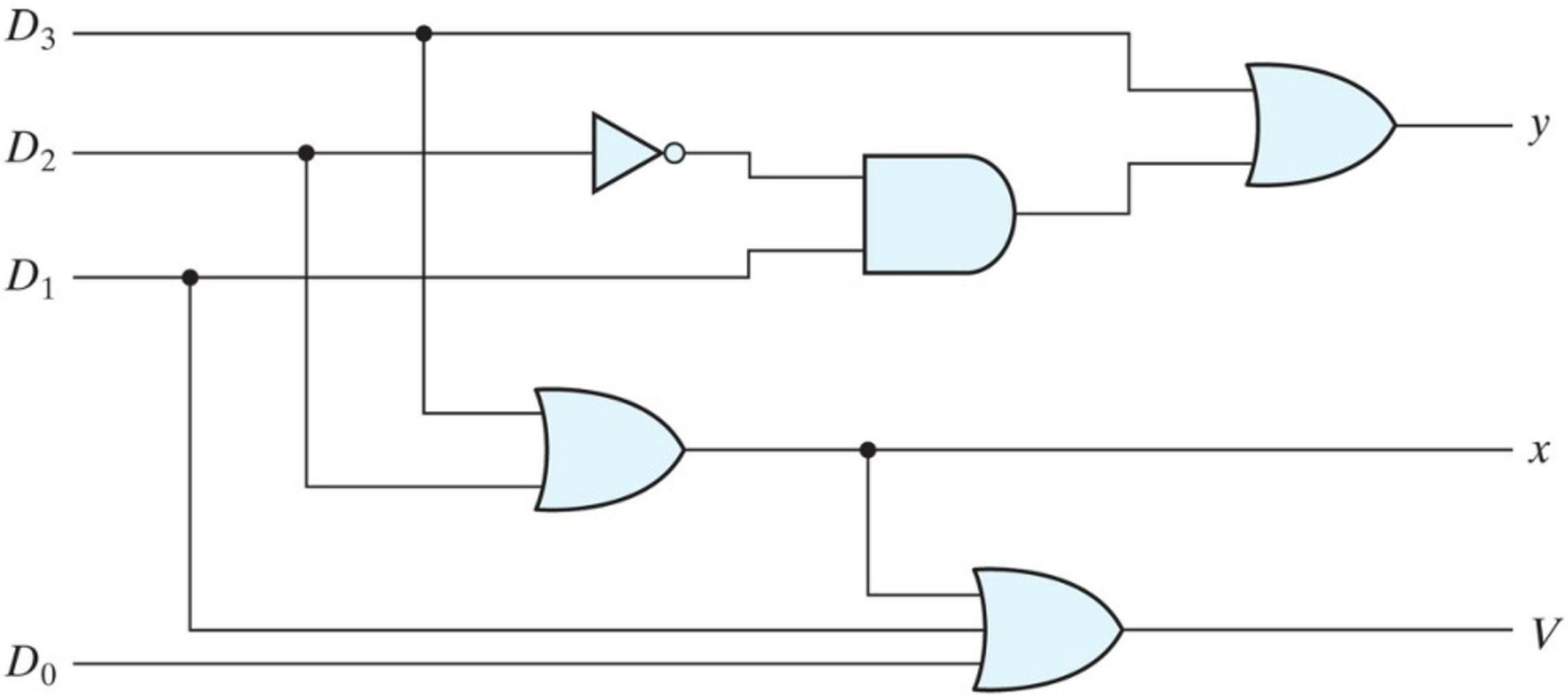


$y = D_3 + D_1D'_2$

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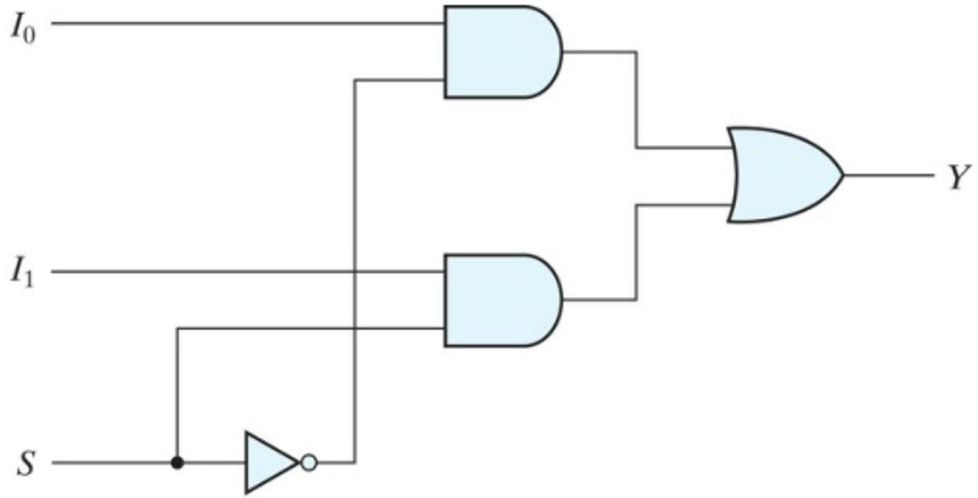
FIGURE 4.23 Four-input priority encoder



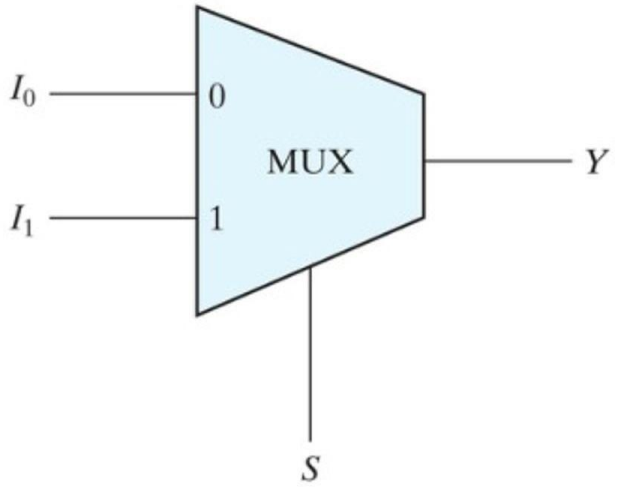
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FIGURE 4.24 Two-to-one-line multiplexer



(a) Logic diagram

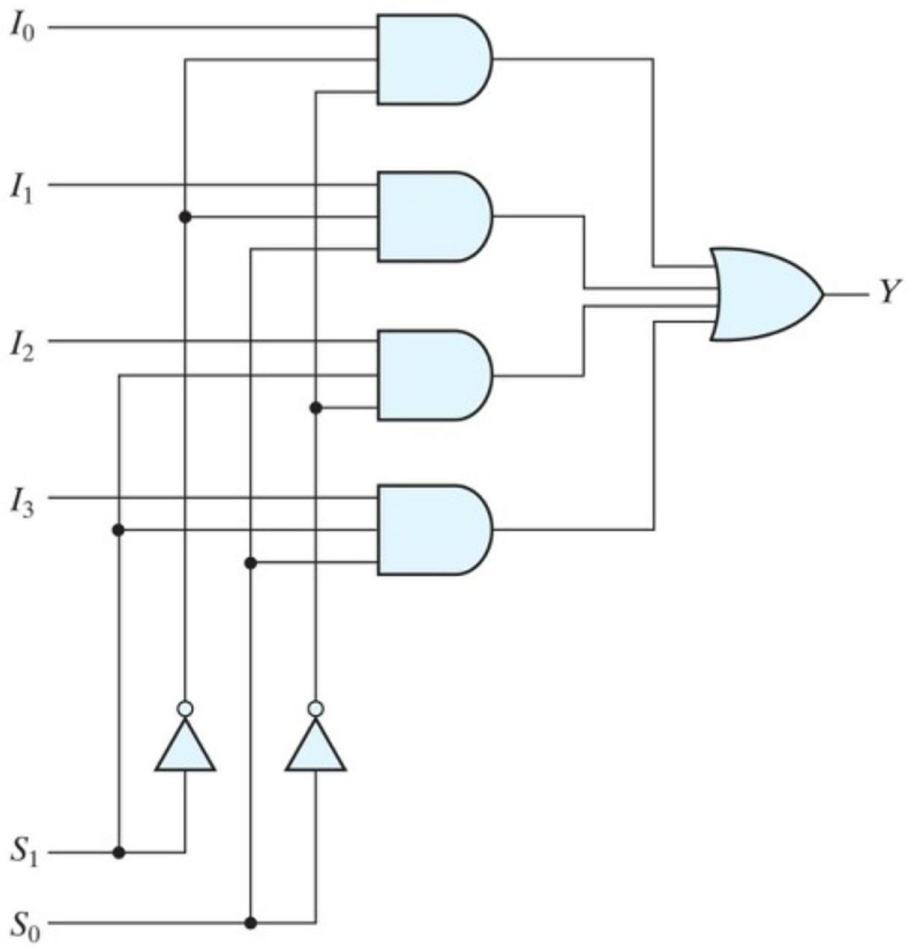


(b) Block diagram

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FIGURE 4.25 Four-to-one-line multiplexer



(a) Logic diagram

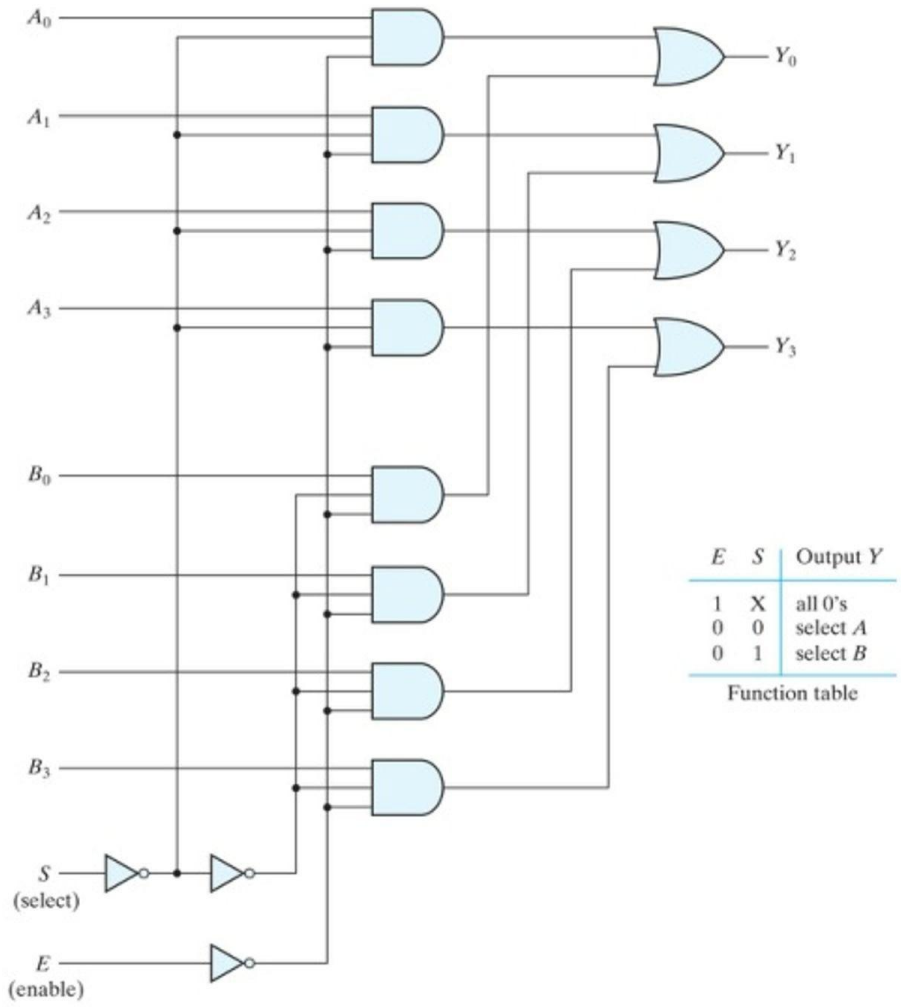
S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

(b) Function table

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FIGURE 4.26 Quadruple two-to-one-line multiplexer



E	S	Output Y
1	X	all 0's
0	0	select A
0	1	select B

Function table

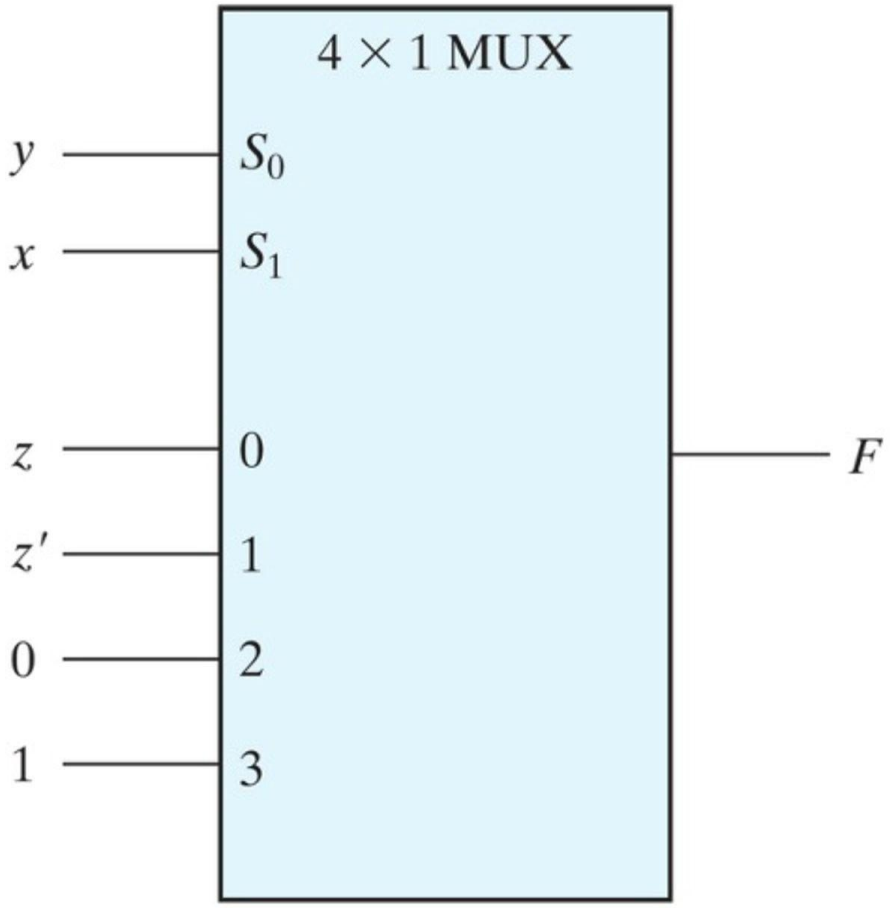
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FIGURE 4.27 Implementing a Boolean function with a multiplexer

x	y	z	F	
0	0	0	0	$F = z$
0	0	1	1	
0	1	0	1	$F = z'$
0	1	1	0	
1	0	0	0	$F = 0$
1	0	1	0	
1	1	0	1	$F = 1$
1	1	1	1	

(a) Truth table



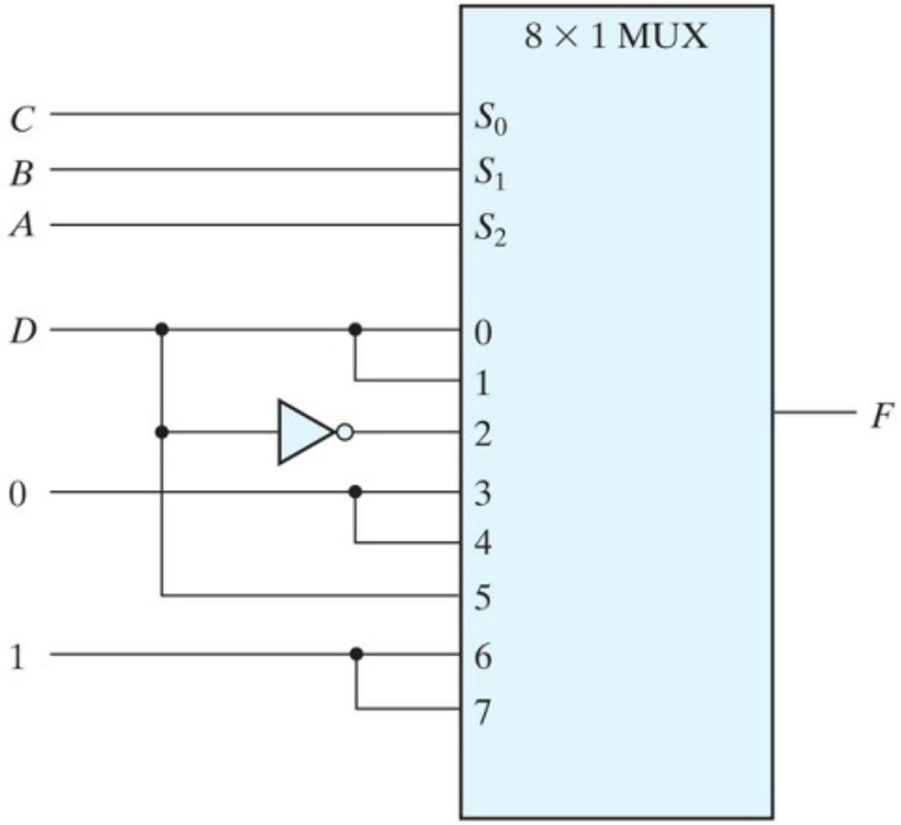
(b) Multiplexer implementation

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FIGURE 4.28 Implementing a four-input function with a multiplexer

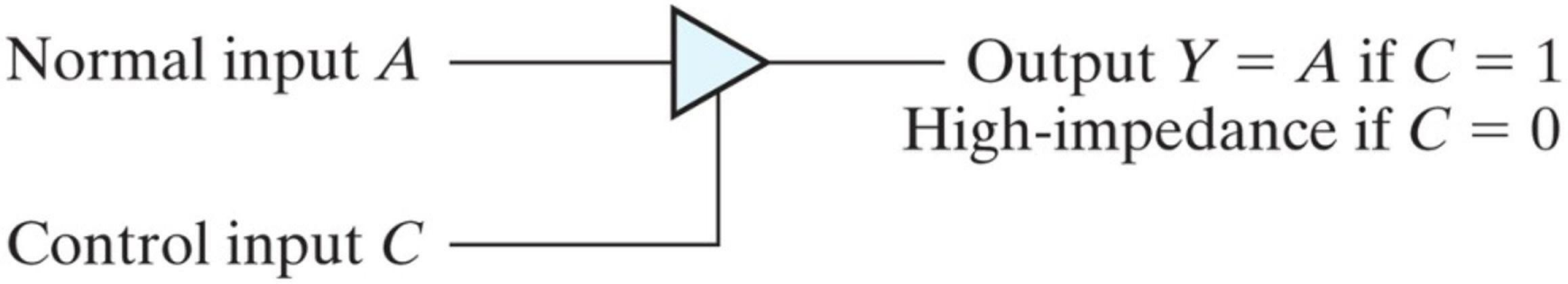
A	B	C	D	F	
0	0	0	0	0	$F = D$
0	0	0	1	1	
0	0	1	0	0	$F = D$
0	0	1	1	1	
0	1	0	0	1	$F = D'$
0	1	0	1	0	
0	1	1	0	0	$F = 0$
0	1	1	1	0	
1	0	0	0	0	$F = 0$
1	0	0	1	0	
1	0	1	0	0	$F = D$
1	0	1	1	1	
1	1	0	0	1	$F = 1$
1	1	0	1	1	
1	1	1	0	1	$F = 1$
1	1	1	1	1	



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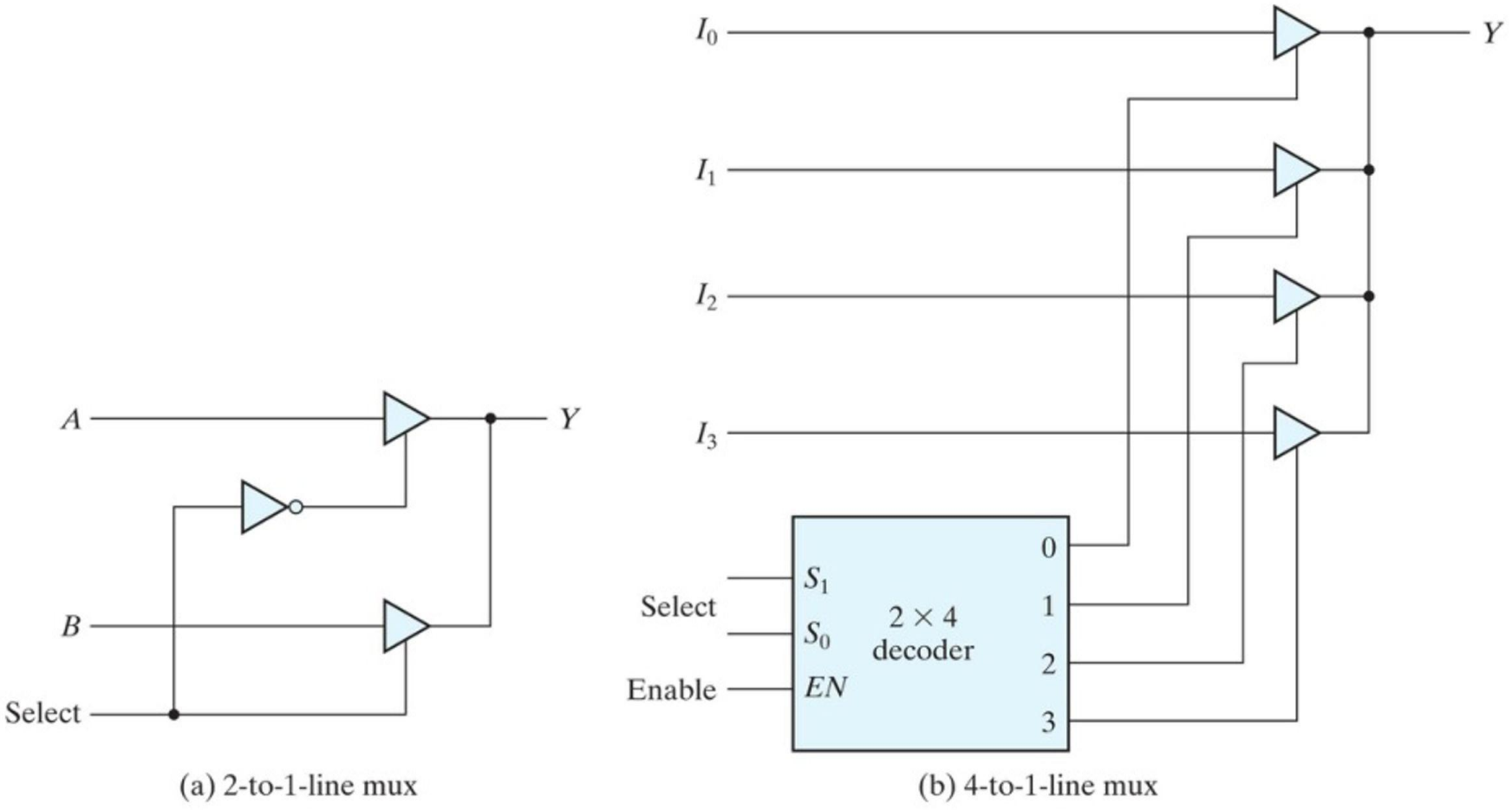
FIGURE 4.29 Graphic symbol for a three-state buffer



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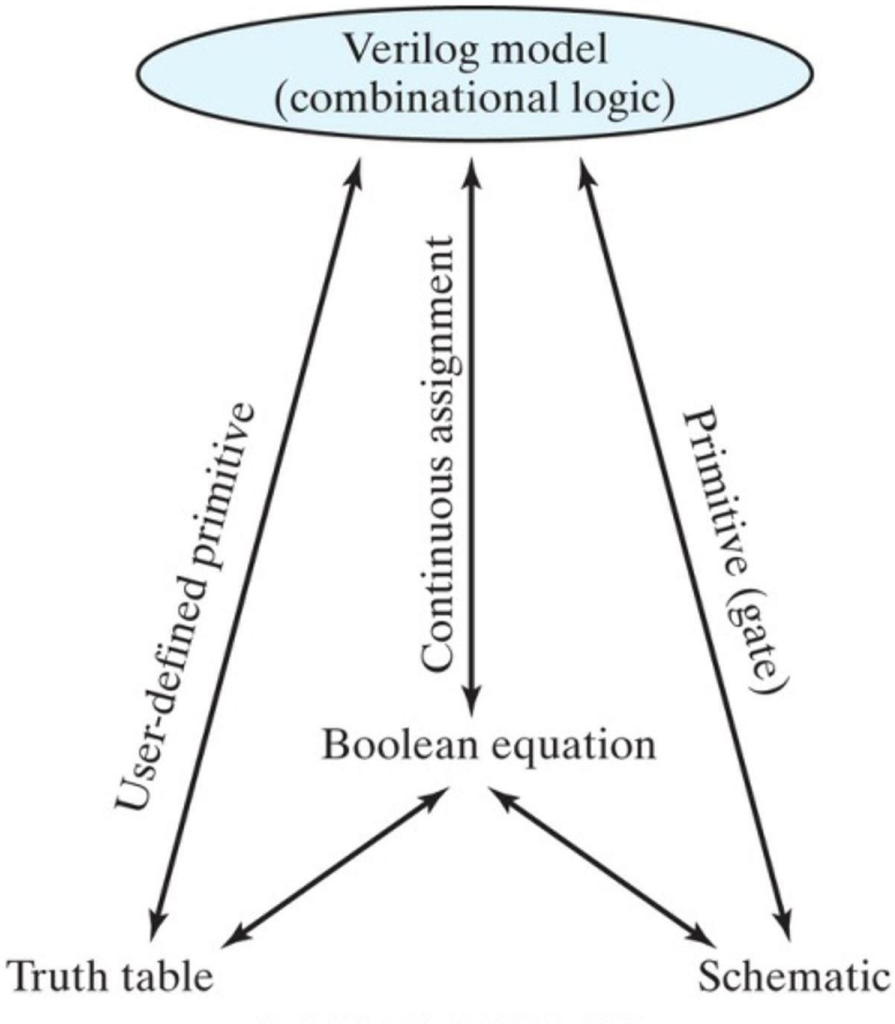
FIGURE 4.30 Multiplexers with three-state gates



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FIGURE 4.31 Relationship of Verilog constructs to truth tables, Boolean equations, and schematics



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Table 4.9 Truth Table for Predefined Primitive Gates

Table 4.9
Truth Table for Predefined Primitive Gates

and	0	1	x	z
0	0	0	0	0
1	0	1	x	x
x	0	x	x	x
z	0	x	x	x

or	0	1	x	z
0	0	1	x	x
1	1	1	1	1
x	x	1	x	x
z	x	1	x	x

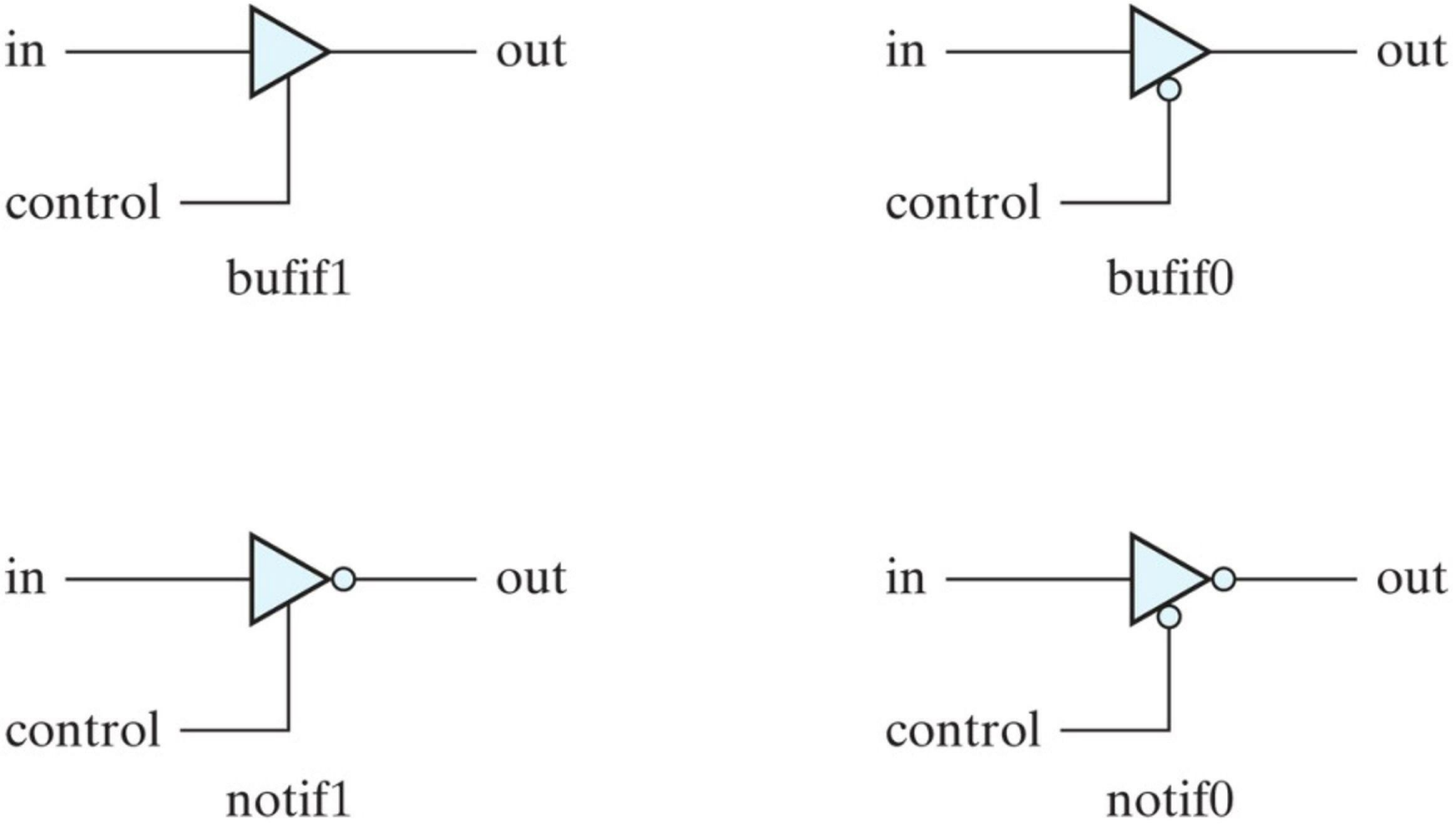
xor	0	1	x	z
0	0	1	x	x
1	1	0	x	x
x	x	x	x	x
z	x	x	x	x

not	input	output
	0	1
	1	0
	x	x
	z	x

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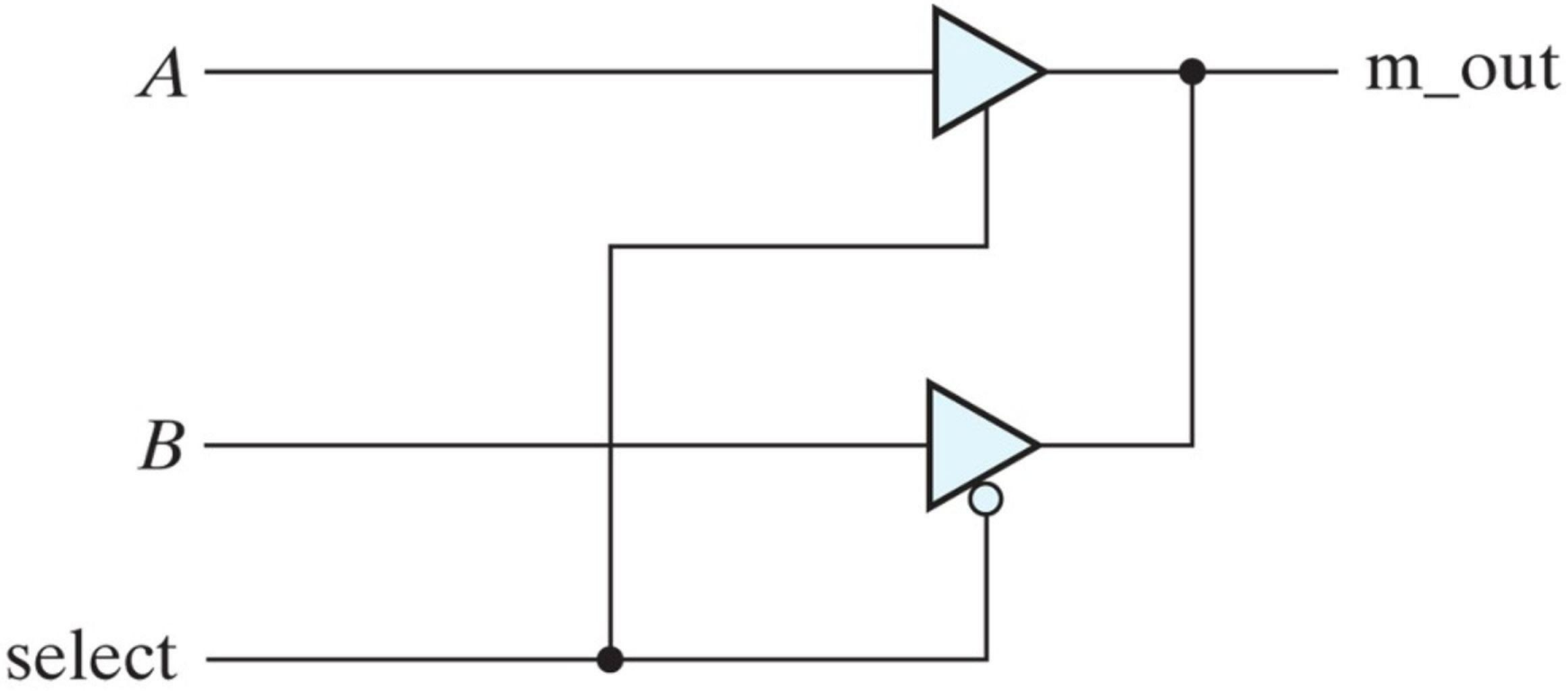
FIGURE 4.32 Three-state gates



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FIGURE 4.33 Two-to-one-line multiplexer with three-state buffers



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Table 4.10 Some Verilog HDL Operators

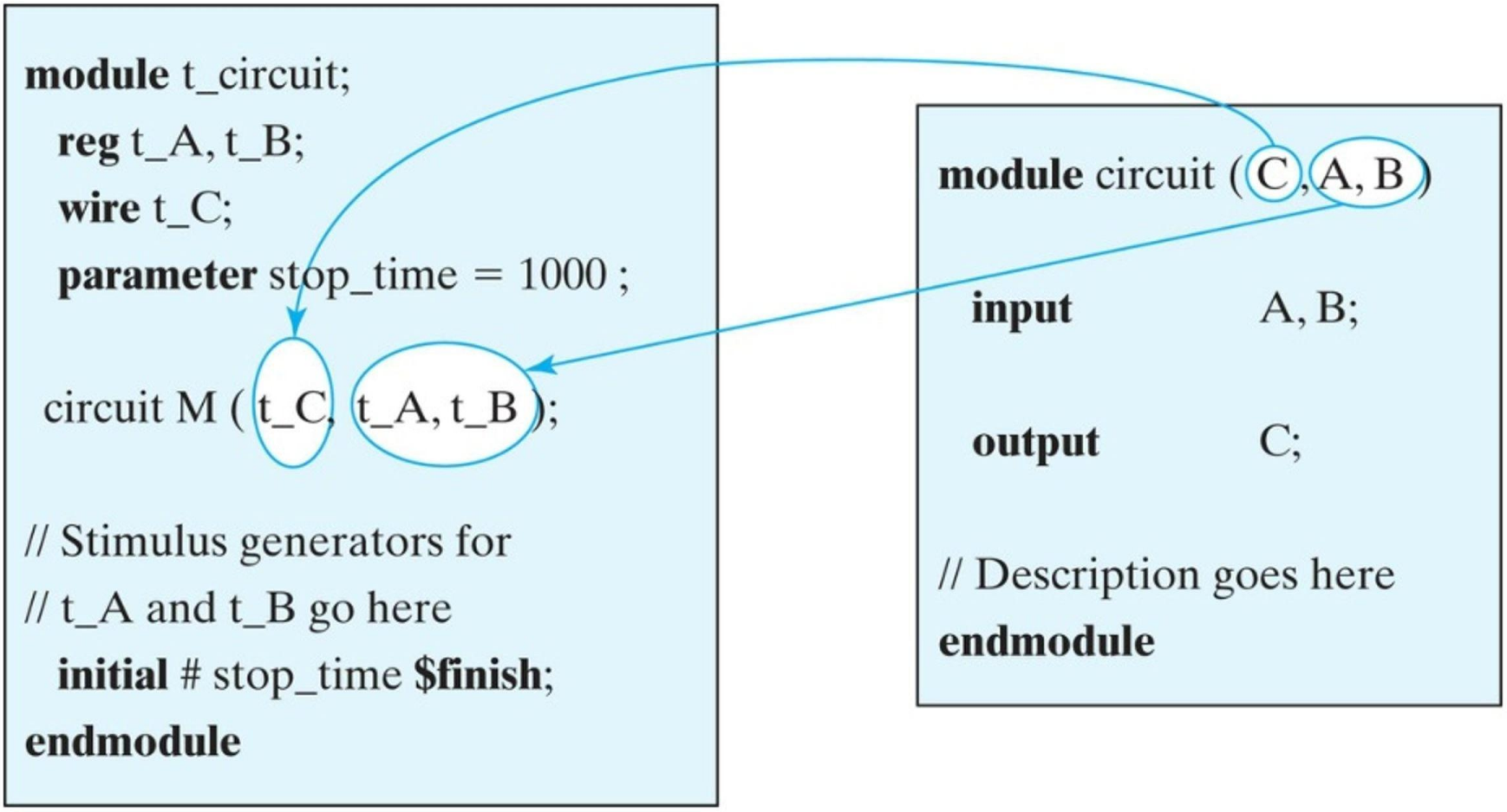
Table 4.10
Some Verilog HDL Operators

Symbol	Operation	Symbol	Operation
+	binary addition		
−	binary subtraction		
&	bitwise AND	&&	logical AND
|	bitwise OR	||	logical OR
^	bitwise XOR		
~	bitwise NOT	!	logical NOT
= =	equality		
>	greater than		
<	less than		
{}	concatenation		
?:	conditional		

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FIGURE 4.34 Interaction between stimulus and design modules



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